

## AGRICULTURE FOR PRODUCING BOTH FUEL AND FOOD: OPTIMISM AND PRUDENCE FOR INDIA

Yogesh Bhatt, Nilabja Ghosh and M. Rajeshwor\*

### ABSTRACT

*This paper considers the prospect of embracing biofuel as a reasonable energy source for the future of the world where dependence on fossil fuel faces the threat of exhaustion and environmental adversities. In the backdrop of experiences of other countries of producing and consuming ethanol and biodiesel based on farm-generated feedstock, the paper weighs India's option of switching to biofuel for sustainable growth and agricultural development against possible trade-offs like land diversion from food and possible but less discussed environmental implications.*

**Keywords:** Crude oil, biofuel, GHG emission, food security  
**JEL Classification:** Q16, Q18, Q20, Q42, Q54

### INTRODUCTION

Biofuel is increasingly seen as a cheaper option for transportation over petrol, diesel and other fossil fuels. This source of energy merits special consideration in countries like India that have to incur large import bills on crude petrol. Worldwide, biofuel has received attention also due to its favorable environmental implications. It is recognized to be cleaner than conventional fuels measured by its emissions. Substitution of conventional hydrocarbon by biofuel could be a powerful way of reducing Greenhouse gas (GHG) emissions to combat global warming and climate change.

As an emerging economy India faces the challenge of meeting energy demands for powering high growth which should also be

sustainable while complying with international norms and minimizing natural hazards that global warming may cause. Given that some of the crops grown in India namely, maize, soyabean and sugarcane are potential feedstock for biofuel production, substitution of conventional fuels by biofuel as a public policy is an option. This policy can potentially create new markets for agro-commodities that in turn will generate more employment in agriculture and agro-processing and enhance farm incomes. Yet, the replacement in the energy sector will go hand in hand with substitutions within agriculture as land will be diverted from food to fuel and traditional emissions of fuel use will give way to emissions generated elsewhere. Thus production of fuel in agriculture is not without cost. Such policy needs caution and deliberation.

India, though deficient in natural fossil fuel resources has not yet firmly set foot on the path of producing biofuel in agriculture. Although producing foodgrains at comfortable levels now, India is still not relieved

\*Research Assistant, Associate Professor and Research Analyst, Institute of Economic Growth, University of Delhi Enclave (North Campus), Delhi-110007

Email: bhatt.yogesh0@gmail.com

of the challenges of food insecurity. Like crude oil, edible oils are also major import items in India. Besides, there are also misgivings about biofuel's potentials for reducing GHG emissions. In other words the biofuel sector is still nascent in India and calls for research, even if based on evidences at the international level, for future planning. This paper gives an overview of the emergence of agriculture generated biofuel and, drawing lessons from other countries, discusses the prospects and problems facing India.

### The Energy crisis

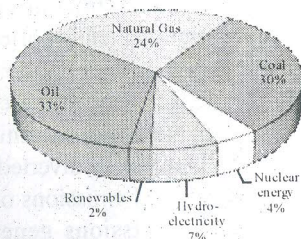
Fossil fuels fulfill about 87 percent of world's total energy requirements but they are exhaustible and fast depleting resources. The most important source of energy is petroleum oil with a share of 33 percent, followed by coal and natural gas with shares of 30 percent and 24 percent, respectively (Figure 1). It is supposed that at the present rate of exploitation oil reserves are enough for less than a century, natural gas will last not much more than six decades and there is enough coal for about 133 years (Chen, 2011). Undoubtedly, human race faces the compulsion of planning a transition to sustainable energy sources even when many

countries are faced with the difficult imperative to develop, grow and move from traditional to modern methods of production.

Crude oil often associated also with the location of natural gas is found only in limited pockets of the world. Middle Eastern countries together have a share of 49 percent of both natural gas and crude oil. With this concentration of natural advantages, oil producing countries came together forming the Organization of the Petroleum Exporting Countries (OPEC)<sup>1</sup> in 1960 and leading to a steep rise in energy prices in early 1970s. The crude oil price rose in the 1960s and 1970s and subsequently also in the 1990s when emerging countries like India, China, Malaysia and Indonesia embarked on economic reforms with a development agenda that transformed the type of energy used from traditional (human and animal) to fossil fuel based. Further the volatility of the price with its reactions to sudden international events and perceptions (Table 1) of uncertainty in which a world dependent on fossil fuel has been living. The geo-political events unfolding in the wake of the formation of the OPEC changed the human perspective of the way the world development was taking place in the aftermath of the industrial revolution.

Soon, research in various disciplines started placing increased focus on other options variously described by *alternative*, *non-conventional* and *renewable* energy. Solar, hydrological, tidal, wind, nuclear and biological sources received attention in research, policy and industry. Although the average share of fossil fuel in world energy consumption came down from 83.8 to 80.7 percent during the period 1971-80 to 2001-10 (World Bank, IEA), the share largely stagnated at that level in the recent decades. Also, the declining trend was manifest only in a handful of high energy consuming nation groups such as USA, Canada and EU. Hardly evident in the developing countries, the imperative for the shift loomed as a challenge in international relations. The global energy demand is

Figure 1: Sources of world energy consumption by type, 2013



Source: BP, 2014

<sup>1</sup>OPEC's mandate is to "coordinate and unify the petroleum policies" of its members and to "ensure the stabilization of oil markets in order to secure an efficient, economic and regular supply of petroleum to consumers, a steady income to producers and a fair return on capital for those investing in the petroleum industry

**Table 1: Crude oil prices during 1861-2013**

| Period  | Events                                                   | End period<br>price @ US \$<br>2013 | Average<br>price @ US<br>\$ 2013 | Periodic<br>growth rate<br>(%) |
|---------|----------------------------------------------------------|-------------------------------------|----------------------------------|--------------------------------|
| 1861-69 | Pennsylvanian oil boom                                   | 63.4                                | 60.2                             | 13.9                           |
| 1870-79 |                                                          | 21.4                                | 44.6                             | -10.3                          |
| 1880-89 | Russian oil exports began                                | 24.3                                | 21.3                             | 0.5                            |
| 1890-99 | Sumatra production began                                 | 36.0                                | 25.0                             | 9.7                            |
| 1900-09 | Discovery of Spindletop, Texas                           | 18.1                                | 20.5                             | -4.5                           |
| 1910-19 | First World War                                          | 27.1                                | 22.0                             | 8.0                            |
| 1920-29 | Fears of shortage in US, Growth of Venezuelan production | 17.3                                | 20.0                             | -3.2                           |
| 1930-39 | East Texas field discovered                              | 17.1                                | 16.0                             | 6.6                            |
| 1940-49 | Post-war reconstruction                                  | 17.4                                | 16.7                             | 0.8                            |
| 1950-59 | Loss of Iranian supplies, Suez crisis                    | 16.6                                | 16.2                             | 0.9                            |
| 1960-69 | Formation of OPEC                                        | 11.4                                | 13.0                             | -2.4                           |
| 1970-79 | Yom Kippur war, Iranian revolution                       | 101.4                               | 45.1                             | 27.0                           |
| 1980-89 | Netback pricing introduced                               | 34.2                                | 55.2                             | -13.8                          |
| 1990-99 | Iraq invaded Kuwait, Asian financial crisis              | 25.1                                | 27.4                             | -4.4                           |
| 2000-09 | Invasion of Iraq                                         | 67.0                                | 60.2                             | 15.3                           |
| 2010-13 | Arab Spring                                              | 108.7                               | 105.5                            | 7.5                            |

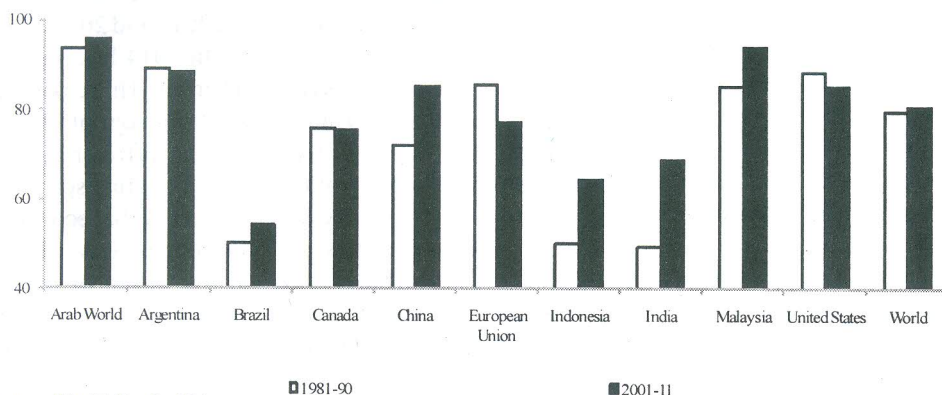
Source: BP, 2014

expected to increase by one third from 2011 to 2035 but the share of fossil fuels in the world's energy is expected to 76 percent (IEA, 2012 and 2013)

For the emerging economies already constrained by their access to energy, a further challenge came from the Kyoto Protocol (KP) signed in Japan and adopted in 1997 by which 37 industrialized countries committed themselves to reductions of GHG emissions from economic activities. The world was slow in waking up to the fact that the GHG concentration, attributed especially to the industrial revolution based on intensive use

of fossil fuel generated energy, was warming up the earth in an irreversible way although systematic recording of weather data was making the inference easier over time. In 2005, Exxon Mobil's report acknowledged that the *accumulation of greenhouse gases* poses risks for society and ecosystem that justify actions (Ghosh, 2011) and paved the way to international conventions and protocols. The KP went through a path which was far from smooth with US refraining from ratifying the treaty and India, a large developing country joining only in 2002 (Wikipedia).

Indeed the shift in paradigm and any

**Figure 2: Fossil fuel as percent of total fuel consumption**

Source : World Bank, IEA

**Table 2: Energy use efficiency in major economies, 2010**

| Emitter countries<br>(Rank wise) | CO <sub>2</sub><br>emissions<br>(Million) | CO <sub>2</sub><br>emissions<br>intensity | Energy use<br>(Million) | Energy use<br>intensity | (Tonnes)<br>Growth rate (%) (2010 over 2000) |                              |
|----------------------------------|-------------------------------------------|-------------------------------------------|-------------------------|-------------------------|----------------------------------------------|------------------------------|
|                                  |                                           |                                           |                         |                         | Co <sub>2</sub> Emission                     | Co <sub>2</sub> Emission per |
|                                  |                                           |                                           |                         |                         |                                              | Capita                       |
| China                            | 8286.9                                    | 1.4                                       | 2516.7                  | 0.4                     | 143.4                                        | 129.7                        |
| USA                              | 5433.1                                    | 0.4                                       | 2215.5                  | 0.1                     | -4.9                                         | -13.3                        |
| EU                               | 3709.8                                    | 0.2                                       | 1724.3                  | 0.1                     | -5.2                                         | -8.3                         |
| India                            | 2008.8                                    | 1.2                                       | 723.7                   | 0.4                     | 69.3                                         | 46.3                         |
| Russian F.                       | 1740.8                                    | 1.1                                       | 702.3                   | 0.5                     | 11.7                                         | 15                           |
| Japan                            | 1170.7                                    | 0.2                                       | 499.1                   | 0.1                     | -4                                           | -4.4                         |
| Germany                          | 745.4                                     | 0.2                                       | 329.8                   | 0.1                     | -10.2                                        | -9.7                         |
| Iran                             | 571.6                                     | 1.4                                       | 210.7                   | 0.5                     | 53.4                                         | 35.8                         |
| Korea Rep.                       | 567.6                                     | 0.5                                       | 250                     | 0.2                     | 26.8                                         | 20.6                         |
| Canada                           | 499.1                                     | 0.3                                       | 251                     | 0.2                     | -6.6                                         | -15.5                        |
| World                            | 33615.4                                   | 0.5                                       | 12515.7                 | 0.2                     | 35.5                                         | 20.1                         |

Source: World Bank

At 000' US\$ GDP

Energy use: Oil equivalent

compromise on fuel consumption can be hard decisions for all poorer and developing countries as also a constraint on industrialized nations used to earlier privileges. The KP however allows for several *flexible mechanisms* such as International Emission Trading (IET), clean development mechanism (CDM) and joint implementation (JI) to enable industrialized countries to meet their GHG emission limitations by purchasing reduction credits from elsewhere.

Financial exchanges are encouraged to facilitate these operations and to be linked to the conventional financial instrument trading markets. The perusal of Table 2 indicates that countries like China, India, Russian Federation and Iran are less efficient in energy use and emission intensity compared to USA, EU, Japan, Germany and emerging economies. China, India, Iran and Russia increased per capita emissions in the 2000s decade while USA and EU reduced them. The situation can be ameliorated if the lagging countries can catch up with efficient technologies and switch to cleaner fuels.

With the new realization, not surprisingly in the most recent years 2011-13 consumption of fossil fuel increased only at a rate by 1.9 percent per year compared to 2.7 percent during the period 2001-10 (BP, 2014) though

fossil fuel's share in total energy consumption moved up from 80 percent to 81.2 percent between in 2001 and 2011 (World Bank, IEA). CO<sub>2</sub> emission has increased by 11.6 percent during period 1990 and 2000 to 35.5 percent during period 2000 to 2010 enhancing the apprehension of irreversible global warming (Table 2).

At the per capita level the emission however fell by 13 percent in US, eight percent in EU, 16 percent in Canada, 10 percent in Germany during 2010 compared to 2000 but in China and India it went up by 130 percent and 46 percent, respectively, during the period (World Bank). Brent crude price grew at 14 percent annually in the decade 2001-10 to reach 79.5 US\$ per barrel from 24.4 US\$ per barrel but came down between 2011 and 2013 by 1.2 percent to 108.7 US\$. In 2014 December it stands at a mere \$60 a barrel!. This change in global tendencies could be a sign of success at achieving energy use efficiency and availability of fuel options but this softening of crude price leads to other global economic complexities and challenges as well. Oil exporting countries face the prospect of defaults creating a threat for the world capital market and India's economy is hardly insulated from the global economic disturbances (Economic Times, 2014).

## Biofuel as an energy source in the World's future

Biofuel, a comprehensive term, is an alternative to conventional fuel that can help in mitigating the energy threats. Generated from farmlands and forest lands devoted to plant-life its production is a regenerative process. Ethanol and bio-diesel are two important primary forms of biofuel. US and Brazil lead in the production and consumption of ethanol while EU and US lead in the production and consumption of biodiesel (Table 5 and 6). Organic materials from agriculture and forest products, directly convertible to combustible fuels, are considered appealing because they generate cleaner emissions besides being non-hazardous, renewable and affordable.

Ethanol's first use was to power an engine in 1826 and in 1876, Nicolaus Otto, the inventor of the modern four-cycle internal combustion engine, used ethanol to power an early engine. In the 1850s it was used as a lighting fuel but its use was curtailed when it was taxed as liquor to help pay for the Civil War. Ethanol use as a fuel continued after the tax was repealed, and fueled Henry Ford's Model T in 1908. The first time ethanol was blended with gasoline was in the 1920s and 1930s (ag.ndsu.edu). In the period of World War II, the high demand for biofuel was due to its increased use as an alternative for imported fuel. Serious shortage of fuel in Germany accelerated the use of gasoline along with alcohol that was derived from potatoes. Further UK explored the concept of grain alcohol mixed with petrol (biofuel.org.uk). In 1990 United States Environmental Protection Agency released the report-*Analysis of the Economic and Environmental Effects of Ethanol as an Automotive Fuel*. It explores the properties and conducted economic analysis of ethanol as a fuel (ww.epa.gov). During 1980 USA's ethanol production was 175 Million Gallon (MG) which crossed the 1000 MG mark in 1993 and 2000 MG mark in 2002. A sharp acceleration was witnessed in US between 2005 and 2013 when production

reached 13000 MG from 3900 MG (Renewable Fuel Association). Brazil did not show similar performance in the same period rising by 24.5 percent in 2013 from 5020 MG in 2007. In response to the dramatic rise in imported petroleum prices in 1973 in Brazil the first ethanol program (ProAlcool) was launched in 1975. Since then the production has increased by more than fifty folds till 2010 (Meyer *et al.*, 2012 and Goldemberg *et al.*, 2004). India produced about 53 MG of ethanol in 2007 and reached 545 MG in 2013 with a share of nearly two percent in world map.

Production and use of biofuels started in several European countries during 1990s (Thuijl and Deurwaarder, 2006). Germany and France are the major consumers in the EU region for biodiesel and ethanol, and EU itself is a major importer of biodiesel and ethanol due to high demand. Input use for producing biofuel reflects regional variation based on the availability and domestic use pattern of the feedstock. Worldwide major feedstock used for the production of ethanol and biodiesel are given in Table 3.

Ethanol constitutes more than 80 percent of global biofuel production which is geographically highly concentrated, Brazil and US together contributing about 86 percent of world ethanol production by 2011 (FAPRI Outlook, 2012). The other producers are EU,

**Table 3: Sources of biofuels in different producer countries**

| Countries | Ethanol sources                                                   | Biodiesel sources                                         |
|-----------|-------------------------------------------------------------------|-----------------------------------------------------------|
| U.S.      | Corn (almost 100%), Corn Stover, Switch grass                     | Soybean oil (99%), Canola oil (1%), Other Fats and oils   |
| Brazil    | Sugarcane (almost 100%)                                           | Soybean oil (almost 100%)                                 |
| E.U.      | Wheat (26%), Barley (4%), Corn (9%), Sugar Beet (58%), Other (2%) | Rapeseed oil (80%), Soybean oil (16%), Sunflower oil (4%) |
| India     | Sugarcane (major), Maize                                          | Jatropha (major)                                          |
| Canada    | Corn (73%), Wheat (27%)                                           |                                                           |
| Argentina |                                                                   | Soybean oil (almost 100%)                                 |
| China     | Corn (70%), Wheat (30%)                                           |                                                           |
| Indonesia |                                                                   | Palm oil (almost 100%)                                    |
| Malaysia  |                                                                   | Palm oil (almost 100%)                                    |

*Note: Share in terms of feedstock used as an input. Source: FAPRI outlooks, Author's computations*

**Table 4: Ethanol production in major producer countries, 2007 versus 2013**

| (Million gallon) |          |          |
|------------------|----------|----------|
| Countries        | 2007     | 2013     |
| USA              | 6521     | 13300    |
| Brazil           | 5019     | 6267     |
| Europe           | 570      | 1371     |
| China            | 486      | 696      |
| Canada           | 211      | 523      |
| Rest of world    | 315      | 1272     |
| World            | 13123.10 | 23429.00 |

Source: US Department of Energy

China, India, Canada and Indonesia. In US nearly all of the ethanol is derived from Maize though switch grass and corn stover are other minor sources. Brazil itself produced about 30percent of the world total ethanol in 2010 (EPE, 2011 and RFA, 2012) and sugarcane is the major feedstock. Wheat, barley, beet and rice are other feedstock in the other producing countries. Ethanol production during two recent time periods is reported in Table 4. Production of biodiesel is concentrated in EU, USA, Brazil and Argentina. More than 95 percent of biodiesel is produced in these countries by 2011. Biodiesel production in Brazil is mainly motivated by domestic consumption needs. Malaysia, China, India, Indonesia are the other minor producers. Indonesia, Malaysia and Argentina produce biodiesel with a focus on exports. The feedstock of biodiesel is Soya bean oil in US, Brazil, EU and Argentina while Rapeseed and Sunflower oils are also used in EU and Canola oil in US. Soya bean oil is the only source in Brazil and Argentina and Palm oil in Indonesia and Malaysia (Table 3). In India biodiesel is produced only from Jatropha, a forest product and ethanol mostly from Sugarcane. Maize, wheat and rice are also used at very low proportion. India's share in world production of ethanol is about two percent and that of biodiesel is negligible. A huge burden of edible oil import due to domestic demand restricts India from using edible oilseeds to produce biodiesel (National Policy on Biofuels, 2009). Biofuel use is expected to increase by more

than three folds from 1.3 million barrels of oil equivalent per day (mboe/d) in 2010 to 4.5 mboe/d in 2035, driven primarily by blending mandates. Supply of ethanol is expected to rise from 1 mboe/d to 3.4 mboe/d during this period. About 37 percent of road transport fuel demand in Brazil, 19 percent in the US and 16percent in the EU is expected to be met by biofuels (IEA, 2012).

Biofuel was first promoted in EU with the introduction of *The Directive on the promotion of the use of biofuels or other renewable fuels for transport (2003/30/EC)* (Amezaga *et al.*, 2010). In 2003 the indicative target for biofuel use in the total volume of fuel consumed in EU was set at two percent by 2005 and the target for 2010 was revised to 5.75 percent. Biofuel use increased more than two folds between 2003 and 2005 (The German Marshall Fund of the United States, 2007) but at 1.4 percent in 2005, it fell short of the two percent target (Schnepf, 2006). In US a proposal to raise the blending target to a high level (approximately 15 percent) is highly debated ([www.worldwatch.org](http://www.worldwatch.org)). Currently, U.S. Environmental Protection Agency (EPA) controls the nation's biofuel standard. Biofuel Blending norms are controversial for their impact on biofuel industry, environmental concerns and other macroeconomic aspects.

An estimated global compound annual growth of biofuel production by six percent in 2013-2023 ([www.greencarcongress.com](http://www.greencarcongress.com)) manifests the projected rise of the biofuel. The emergence of biofuel has generated options

**Table 5: Ethanol: Major producer and consumers in 2011**

| (Million gallon) |            |             |
|------------------|------------|-------------|
| Countries        | Production | Consumption |
| USA              | 13,823     | 13,118      |
| Brazil           | 6,456      | 6,232       |
| EU               | 1,614      | 1,984       |
| China            | 725        | 724         |
| India            | 511        | 531         |
| Canada           | 357        | 515         |
| Indonesia        | 172        | 94          |

Source: FAPRI outlooks

**Table 6: Biodiesel: Major producer and consumers in 2011**

| Countries | (Million gallon) |             |
|-----------|------------------|-------------|
|           | Production       | Consumption |
| EU        | 3,079            | 3,731       |
| USA       | 899              | 960         |
| Brazil    | 719              | 715         |
| Argentina | 676              | 251         |
| Malaysia  | 15               | 1           |

Source: FAPRI outlooks

both in the energy market and in agriculture. A remarkable competitive spiraling also became evident in the fuel market. World ethanol prices increased from US\$ 1.72 per gallon in 2006 to US\$ 3.30 per gallon till 2011 (FAPRI Outlook, 2012) but in recent time it decreased to 2.05 US\$ per gallon in December 2014 with the average price being US\$ 2.5 per gallon during 2013 to 2014 (www.tradingeconomics.com). In Brazil the increase of ethanol production and consumption by 0.9 and 7.7 percent respectively in 2009 translated to diversion of 333 million tonnes of sugarcane for the purpose from traditional uses.

All this shows that the shift to biofuel triggers competition affecting prices of different fuels, different countries and between different agro-forest based products. While such competitive forces are likely to reveal themselves in coming years as the shift intensifies, some early indications are already visible also in the tendencies and shifts seen in agriculture in the period starting 1995 when biofuel production started gaining ground. A simple trend analysis conducted on acreages under different crops in the biofuel producing countries using data reported by Food and Agriculture Organization (FAO) for the period 1975 to 2008 showed improvement in growth rates of Maize acreage in corn based ethanol producing USA even when it is negative in EU, China, India and Canada, for Sugarcane in sugarcane based ethanol producing country Brazil and also in India while being negative or declining in USA, EU and China (Table 7). No increase in the growth rate of area under the biodiesel feedstock (mustard, rapeseed) is seen in EU but Soyabean acreage has

**Table 7: Worldwide trends in area under the crops used as a feedstock for ethanol and biodiesel production in major producer countries/region**

| Particulars      | USA       | Brazil   | EU        | China    | India     | Canada        | World     |
|------------------|-----------|----------|-----------|----------|-----------|---------------|-----------|
| <i>Ethanol</i>   |           |          |           |          |           |               |           |
| <i>Maize</i>     |           |          |           |          |           |               |           |
| 1975-2008        | 0.003     |          | 0.0004*** | -0.01*** | 0.004***  | 0.011***      | 0.007***  |
| 1995-2008        | 0.01**    |          | -0.003*** | -0.01*** | -0.001*** | 0.01          | 0.01***   |
| <i>Sugarcane</i> |           |          |           |          |           |               |           |
| 1975-2008        | 0.009***  | 0.03***  | -0.097*** | 0.03     | 0.02***   |               | 0.02      |
| 1995-2008        | -0.004*** | 0.04     | -0.19***  | 0.028*** | 0.02***   |               | 0.01      |
| <i>Wheat</i>     |           |          |           |          |           |               |           |
| 1975-2008        | -0.01***  |          | 0.005     | -0.01*** | 0.009     | -0.007        | -0.003*** |
| 1995-2008        | -0.02***  |          | -0.0001   | -0.024** | 0.005     | -0.02         | -0.002    |
| <i>Biodiesel</i> |           |          |           |          |           |               |           |
| <i>Rapeseed</i>  |           |          |           |          |           |               |           |
| 1975-2008        | 0.17***   |          | 0.05***   |          | 0.02***   |               | 0.04***   |
| 1995-2008        | 0.05***   |          | 0.04***   |          | -0.001    |               | 0.02      |
| <i>Soyabean</i>  |           |          |           |          |           |               |           |
| 1975-2008        | 0.007     | 0.035*** | 0.002**   | 0.009*** | 0.12***   | 0.1*** (Arg)  | 0.025***  |
| 1995-2008        | 0.008     | 0.06***  | -0.035    | 0.013*** | 0.04      | 0.09*** (Arg) | 0.04***   |
| <i>Mustard</i>   |           |          |           |          |           |               |           |
| 1975-2008        | 0.023***  |          | 0.07      |          |           |               | 0.015***  |
| 1995-2008        | 0.047***  |          | 0.02      |          |           |               | -0.009    |

Note: Notation (Arg) for the soyabean crop is taken for Argentina for all three periods as Argentina uses almost 100 of its soyabean feedstock to produce biodiesel. \*\*\*, \*\* indicates significance under 5 and 10 percent, respectively, no star is given for insignificant coefficients. Source: FAO

accelerated in Brazil and China. The area under wheat and its growth rates declined in US. Similar performance is seen in respect of the staple wheat in the other countries, most notably EU and China. It may be noted however that the change in acreage can have many reasons other than biofuel production. Also, since the crops used as feedstock are themselves food items the diversion of the same product for use as biofuel input also serves to curtail food availability.

### Biofuel, Food and Other Trade-offs

That biofuel is a panacea for the fuel problem and the developmental paradox facing the world at large can be a delusion. Many caveats inherent in the optimism have been pointed out by pragmatics and the critics. The most glaring of the possible consequence lies in the food-fuel substitution that can make about this shift.

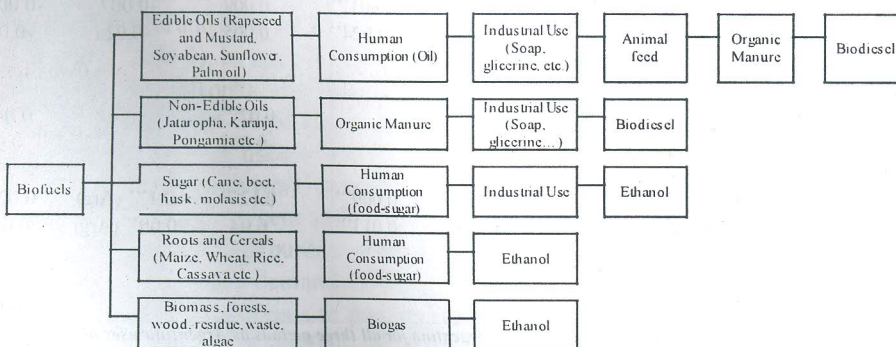
Ethanol and biodiesel are made from crops produced in agriculture. Land is the major factor of production and like the non-renewable forms of energy, land is limited in supply. Moreover in countries with growing population and falling land-man ratio growing of biofuel crops will invariably mean a compromise of other gainful uses. Urbanization and development generally imply demand for land for the purposes of housing, roads and other infrastructure and factories. Development activities have already had their toll on forests and remaining forest cover is

stoutly protected. All this means a substitution between biofuel crops and other crops most of which are food crops. These food crops include cereals that are important as staples besides protein and fat generating crops such as legumes, pulses and oilseeds. Diversion of land to biofuel crops would also compromise production of animal feed and restrain production of animal based food such as meat and dairy products. Moreover the common biofuel crops maize, sugarcane, soya bean are themselves alternative sources of human and animal nutrition and are essential in human diet, and so their use as feedstock can reduce their availability as food or feed (Figure 3).

### Biofuel use in India

For India the challenge of reducing reliance on fossil fuel while meeting a GHG target is a difficult task compared to developed countries. This is clear from the fact that the energy consumption is only 613 kg of Oil equivalent per capita (kgpc) in year 2011 in India compared to China (2029 kgpc), UAE (7407 kgpc), EU (3285 kgpc), Russian Federation (5113 kgpc) and USA (7032 kgpc) (Word Bank). The world energy consumption during same year was 1890 kgpc. India has a population of about 1.25 billion with 269.3 million living under the poverty line in 2011-12, as estimated by Tendulkar committee, using Mixed Reference Period (MRP) (Planning Commission, 2013). Dependence on energy sources such as fire wood and animal power is

Figure 3: Usefulness of biofuel feed stock derived from agriculture



common among the rural people. Modernization of the traditional small enterprises and the large agricultural sector to accelerate industrialization, generate employment and deliver acceptable standards of living to large sections of people is the most important objective of the government. In this endeavor the government confronts the challenge of meeting environmental safeguards that tend to come in conflict with the objective. The consumption of oil has increased from 2.3 million barrels per day in 2001 to 3.7 million barrels per day in 2013 indicates the continuous rise in energy demand, as reflected in Table 8.

For India dealing with the energy crisis is not just meeting the resources required for oil imports. Sustainable development is now also ingrained in India's economic policy as the objective transcending growth imperatives. Achieving development needs with the least possible damage inflicted on the environment appears as a serious socio-political issue. It is only prudent that development plans factor in the possible adversities of global warming on the economic and social lives of people

**Table 8: India's daily oil consumption and consumption share in world**

| Years | (Million gallon)                        |                                       |
|-------|-----------------------------------------|---------------------------------------|
|       | Oil Consumption<br>'000 barrels per day | Share in World Oil<br>Consumption (%) |
| 1965  | 252.6                                   | 0.8                                   |
| 1975  | 477.4                                   | 0.9                                   |
| 1985  | 896.9                                   | 1.5                                   |
| 1995  | 1581.3                                  | 2.2                                   |
| 2001  | 2288.3                                  | 2.9                                   |
| 2002  | 2413.4                                  | 3.1                                   |
| 2003  | 2485.3                                  | 3.1                                   |
| 2004  | 2555.5                                  | 3.1                                   |
| 2005  | 2605.6                                  | 3.1                                   |
| 2006  | 2736.7                                  | 3.2                                   |
| 2007  | 2940.8                                  | 3.4                                   |
| 2008  | 3076.9                                  | 3.6                                   |
| 2009  | 3236.7                                  | 3.8                                   |
| 2010  | 3319.3                                  | 3.8                                   |
| 2011  | 3488.3                                  | 3.9                                   |
| 2012  | 3685.4                                  | 4.1                                   |
| 2013  | 3726.6                                  | 4.1                                   |

Source: BP-Statistical Review of World Energy

because even as growth is important for development, frequent natural disasters, coastal submergence and the pressure imposed on farming practices and calendars by climate change can be costly in terms of budgetary resources and welfare.

Although, the production of biofuel is at early stages in India, in the last few years the policy has however been less than static. Blending petrol with 5 percent of biofuel by 2012, raised to 10 percent in 2017 is made mandatory though there are bans of use of edible oilseeds to produce biodiesel and the sugarcane molasses is used for ethanol production. In order to fulfill 10 percent blending target by the year 2016-17, Government of India will require producing approximately 736.5 million tonnes of sugarcane with area coverage of 10.5 million hectare (Raju *et al.*, 2012). Large scale biofuel production is held in check by apprehensions of scarcity generated of other necessary products. India has been producing ethanol at global map since last decade and the production reached a peak in 2007 (633 million gallon) after which it came down to 511 million gallon in 2011 (FAPRI outlook, 2012) (Table 9). Biodiesel production is negligible and focus is on tree-borne oil and policy for assuring a minimum support price is under consideration. The National Biofuel Policy has a vision of accelerating cultivation, production and use of biofuel but the possible conflicts with other objectives, most notably the national food security is a major concern.

**Table 9: India's ethanol production and consumption**

| Years | (Million gallon) |             |
|-------|------------------|-------------|
|       | Production       | Consumption |
| 2005  | 330.0            | 385.0       |
| 2006  | 501.4            | 432.9       |
| 2007  | 633.4            | 462.3       |
| 2008  | 567.9            | 512.5       |
| 2009  | 283.4            | 470.2       |
| 2010  | 379.1            | 470.2       |
| 2011  | 510.9            | 530.9       |

The adequacy of food production is a serious factor for achieving food security but it is not the only one (Ghosh and Khasnabis, 2006). In India the green revolution has helped to achieve high levels of production of rice and wheat and management of stocks has become a serious challenge. Public procurement, distribution targeting the poorest and the recent legislation on the right to food are policies to enable all people to attain desirable nutrition. However, the achievement on the nutrition front has been less than satisfactory. One factor highlighted by critics is the limited role of food sufficiency triggered by technology in determining food security. In reality, poverty and poor infrastructure both coming in the way, the market has not helped in equitably distributing the produced food for meeting the needs of the people, especially the most vulnerable. Despite large budgetary allocations on subsidies and the occasional experimental innovations the public distribution system has also not adequately served the purpose even though it operates at large administrative costs and deprives the country of other required amenities that call for public investment and administration.

A second factor that also gained recognition is the limited perspective taken of food security that has driven India's food policy. Nutrition is not just intake of calories but other nutrients like proteins, fats and vitamins are also important for a healthy and meaningful life. Incidences of child malnutrition and deficiency diseases have been shown to be high while resistance to other diseases is also undermined by poor nutrition. Thus shortage of pulses, a major source of protein in India also needs to be mitigated. Similarly deficit in oilseed makes India dependent on edible oil imports for fat and protein. Thus, food security in a market economy and especially in a globalized market cannot be equated with food grain production as access to various type of food can come via market provided the purchasing power is present. This is why the poor are most

vulnerable to food insecurity and are likely to be hit hardest by any adverse shock in the food market. Livelihood is thus also an important contributor to food security.

In this background large scale biofuel production inclined with the competitive process of the market raises optimism as well as apprehensions. Market opportunity created by biofuel use for energy can help enhance farm incomes and purchasing power. On the contrary, India hardly has any advantage for producing biodiesel from agriculture and such a move will intensify the scarcity of edible oil. Edible oil is produced from groundnut, rapeseed and soyabean with conversion rates being low in India by international standards. Soya bean, the dominant feedstock in major biofuel producing countries is grown only in a limited belt but in physical measure it has the largest share, 47.4 percent, of India's oilseed production (Ministry of Agriculture, 2012). It is produced almost entirely in three states, approximately 53 percent in Madhya Pradesh followed by 32 percent in Maharashtra and 10 percent in Rajasthan (Ministry of Agriculture, 2012). The import of edible oils in India during October, 2013-November, 2014 is likely to stand at 11.8 million tonnes meeting nearly 65.3 percent of consumption needs (Business Standard, 2014). Unless the productivity of oilseed production is improved significantly accompanied by more efficient pressing technology, the potential for biodiesel production does not exist. Besides oilseeds are also sources of industrial inputs and the emerging processing industry would benefit from greater availability. They are also used for producing animal feed and any diversion for biofuel generation will affect the dairy sector which is also seen to be of emerging significance for livelihood and food security.

Ethanol is produced from maize, other cereals and sugarcane. Maize is a crop grown in various parts of the country and makes little demand for resources. It is not an irrigated crop in many regions and in some states like- Bihar, Andhra Pradesh, Karnataka and West Bengal,

a number of maize crops (Kharif, Rabi and Zayad) can be raised in a year. Maize too has uses other than for purpose of direct consumption as a minor food. Processing industries require maize for producing breakfast cereals, different snacks, corn oil, starch, feed and other products and feed. Rice and wheat are cereal crops with vast political importance but there is a case for considering their surplus as possible feedstock on a case to case basis. Sugarcane is already produced for electricity co-generation and grown in large amounts in several states. While pricing of sugar raises some serious issues in India's political economy, a sound policy on sugarcane production and pricing needs to take account of this potential as well.

Land is a scarce resource in India where the population density in 2013 is 421 people/km<sup>2</sup> of area compared to China (145 people/km<sup>2</sup> of area) and USA (35 people/km<sup>2</sup> of area), (*World Bank*). Irrigation endowment is also limited and unequal. Depending on the land quality and irrigation endowment the land can be cultivated in more than one season though kharif and rabi are the two most dominant crop seasons. Maize and soya bean are grown even under un-irrigated conditions and sugarcane is grown in a long season in states Uttar Pradesh, Maharashtra and Andhra Pradesh. Estimated supply response equations of food crops suggest that biofuel cultivation can mean diversion of land from pulses which are protein crops in India, and oilseeds (Bhatt *et al.*, 2014). Edible oil seed crop groundnut and staple rice will also be affected by large extent by promoting large scale use of sugarcane as a biofuel crop. To protect oilseed production for the purpose of food, bio-diesel in India is restricted to be produced from non-edible oil seeds. Using waste, degraded and marginal lands for the purpose is seen as an appealing option over using farmlands. According to plans plantations of *Jatropha* plant on forest, community and Government owned waste lands will be promoted and no taxes and duties should be levied on bio-diesel. Examples of

biodiesel production initiatives in the country include the establishment of Chhattisgarh Biodiesel Development Authority (CBDA) in 2005 to undertake need based research, Orissa Renewable Energy Development Agency (OREDA) in 2007 as the nodal agency for bio-diesel development in Orissa and similar such establishments in states like: Maharashtra, Uttarakhand, Tamilnadu, Rajasthan, Andhra Pradesh, Madhya Pradesh and Karnataka.

### **Way Ahead**

Biofuel offers a much coveted answer to the human quest for alternative energy sources to overcome the problems of exhaustion and adverse environmental implications of fossil fuel. Biofuel as an option merits serious consideration in India which sets a composite target for high growth, sustainable development and international responsibility. There is a need for serious research on producing and consuming biofuel on larger scale in the medium term future. Given that the experiment with biofuel is yet to take off in India lessons may be learnt from information collected in other parts of the world.

Intuitively, production of ethanol and biodiesel promises to expand market for farm products specifically maize, sugarcane and oilseeds and in the process improve farm incomes, generate employment in industry and agriculture and promote livelihood and welfare. At the same time it will help India to reduce dependence on imports of crude and on limited reserves of coal while enabling the economy to meet compliance standards on pollution and emission. International experiences indicate that use of biofuel draws the markets for energy as well as agricultural products into competitive turmoil and volatility. This means that India has to brace herself for the uncertainty and shifts in tandem with global markets calling for a sound underlying priority fuel and flexible technologies like combined cycle plants and multi-fuel engines. The shadows of food insufficiency linger in the country where monsoon determines farm

performance and deficits in other food products like pulses and oilseeds need to be eliminated making the decision difficult.

Even if oilseeds and other essential food materials are not diverted for the purpose, competition among varied crops for limited land is still likely to result in temporal declines in production of cereals, pulses or oilseeds due to diversion of land. Besides, there are also other issues about the shift from fossil fuel to agriculture based biofuel that need to be taken account. Production of biofuel is not simply cultivation of specific crops but involves transportation of feedstock, drying and processing which would also demand energy and create emission and the net effect may actually fall far short of what is visualized. Shifting from crop based to forestry based production of biofuel as is done in India may affect natural biodiversity of forests and create unknown imbalances. That import of biofuel is an alternative to importing crude cannot be ruled out. The revival of the Global oil-shale industry in recent times and the US legislation permitting the extraction of oil-shale and oil-sands in 2005 generate better prospects for energy outlook. Besides, a next generation of biofuel is under research, development and construction and success in exploring biofuel based on algae and agricultural waste could confront the Indian economy with another turning point. Thus, it is advisable for India continue treading the transition in fuel with caution.

## REFERENCES

- Amezaga, J.M., Boyes, S.L. and Harrison, J.A. 2010. Biofuels policy in the European Union. *7<sup>th</sup> International Biofuels Conference*, February 2010, New Delhi.
- Anonymous. 2014. Dependence on imported edible oil to hit new high Article, *Business Standard*. 19<sup>th</sup> April, 2014 culled from [www.business-standard.com](http://www.business-standard.com).
- Bhatt, Y., Ghosh, N. and Tiwari, N. 2014. The food versus energy security debate over using agriculture to produce bio-fuels: A Time series regression approach. *Paper presented at National Conference on Recent Advances in Statistical and Mathematical Sciences and their Applications*, 2014.
- Chen, F.F. 2011. *An indispensable truth: How Fusion Power Can Save the Planet*. book, Springer, 2011.
- EPE. 2011. Brazilian Energy Balance, Rio de Janeiro, Brazil, Empresa de Pesquisa Energética (EPE) and Brazilian Ministry of Mines and Energy.
- FAPRI Outlook. 2012. Food and agricultural policy research institute *U.S. and World agricultural outlook*. Culled from [www.fapri.iastate.edu](http://www.fapri.iastate.edu).
- Ghosh, N. and Khasnobis, B.G., 2006. *Looking for answers to the food security problem: India under current compulsions*. **WIDER Research Paper, 2006/123**, UNU-WIDER.
- Ghosh, N. 2011. *Role of climate change and agricultural performance*. **Research Report**. IEG, New Delhi.
- Goldemberg, J., Coelho, S.T., Nastari, P.M. and Lucon, O. 2004. Ethanol learning curve-the Brazilian experience. *Biomass and Bioenergy*. 26 (3): 301-304.
- IEA. 2012. World Energy Outlook, 2012.
- IEA. 2013. IEA-Factsheet, 2013.
- IEA, International Energy Agency.
- Meyer, D., Mytelka, L., Press, R., Dall'Oglio, E.L., de Sousa Jr, P.T. and Grubler, A.. 2012. *Brazilian ethanol: Unpacking a success story of energy technology innovation: Historical case studies of energy technology innovation*. In: Chapter 24, *The Global Energy Assessment*. Grubler A., Aguayo, F., Gallagher, K.S., Hekkert, M., Jiang, K., Mytelka, L., Neij, L., Nemet, G. and Wilson, C. Cambridge University Press: Cambridge, UK.
- Ministry of Agriculture. 2012. Directorate of Economics and Statistics, Department of Agriculture and Cooperation, Ministry of Agriculture database. Culled from [www.eands.dacnet.nic.in](http://www.eands.dacnet.nic.in).
- National Policy on Biofuels. 2009. *National Policy on Biofuels, Ministry of New and Renewable Energy*, Government of India.
- Planning Commission. 2013. *Press note on poverty estimates, 2011-12*. Planning Commission, Government of India, July 2013.
- Raju, S.S., Shinoj P., Chand, R., Kumar P. and Msangi, S. 2012. *Biofuels in India: Potential, policy and emerging paradigms*. **Policy Paper 27**, NCAP, New Delhi.
- RFA. 2012. *World fuel ethanol production*. Renewable Fuels Association (RFA). Culled from [www.ethanolrfa.org](http://www.ethanolrfa.org).

- Schnepf, R. 2006. European Union Biofuels Policy and Agriculture: An Overview. (RS22404). [www.biofuel.org.uk](http://www.biofuel.org.uk).  
[www.data.worldbank.org/indicator.ag.ndsu.edu](http://www.data.worldbank.org/indicator.ag.ndsu.edu)  
 The German Marshall Fund of the United States. 2007. European Union Policy on Bioenergy. Washington: GMF. [www.en.wikipedia.org](http://www.en.wikipedia.org).  
[www.worldwatch.org](http://www.worldwatch.org)  
 Thuijl, E.V. and Deurwaarder, E.P. 2006. European biofuel policies in retrospect, Energy research Centre of the Netherlands, ECN-C-06-016.

**Websites quoted**

[www.articles.economictimes.indiatimes.com](http://www.articles.economictimes.indiatimes.com)  
[www.bp.com](http://www.bp.com)

---

*Received: December 31, 2014*  
*Accepted: February 05, 2015*